

High-Yield Dual-Channel Modular Tokamak Battery Architecture: Multi-Physics Derivations & System Design

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September 2026

Abstract

This dossier details the mathematical derivations, magnetic flux coupling, and economic optimization for scaling Fast-Switched Inductive Direct Energy Conversion (DEC) in magnetic confinement fusion. By coupling a 36-sector field matrix to a 32-unit modular tokamak array, the architecture extracts charged particle kinetic energy at 99.70% direct efficiency while funneling neutral thermal loads to a shared 640 MW steam turbine block.

1 Governing Equations & Flux Coupling

The induced Faraday back-EMF ($\mathcal{E}_i$) across sector coil i during localized plasma expansion velocity $v_r = \frac{dr}{dt}$ is governed by:

$$\mathcal{E}_i(t) = -N_{\text{turns}} \cdot B_0 \cdot \left(\frac{2\pi a_{\text{minor}}}{N_{\text{sectors}}} \right) \cdot v_r(t) \quad (1)$$

When connected to load impedance R_{load} , the extraction current I_{ext} and Lenz-law drag force $\mathbf{F}_{\text{drag}}$ are given by:

$$I_{\text{ext}}(t) = \frac{\mathcal{E}_i(t)}{R_{\text{coil}} + R_{\text{load}}}, \quad \mathbf{F}_{\text{drag}} = - \left[\frac{\left(N_{\text{turns}} B_0 \frac{2\pi a}{N_{\text{sectors}}} \right)^2}{R_{\text{coil}} + R_{\text{load}}} \right] v_r(t) \quad (2)$$

2 Linear Programming Plant Optimization

The plant capital expenditure (Z) is minimized subject to net electrical power constraints ($P_{\text{net}} \geq 1200$ MW):

$$\text{Minimize } Z = C_{\text{module}}x_1 + C_{\text{turbine}}x_2 + C_{\text{land}}x_1 \quad (3)$$

$$\text{Subject to: } (P_{\text{DEC}} \cdot x_1) + x_2 - (P_{\text{house_load}} \cdot x_1) \geq 1200 \text{ MW} \quad (4)$$

Solving yields an optimal configuration of $x_1 = 32$ modular units and $x_2 = 640$ MW shared turbine capacity.